

INPUT

M

METAMODEL

DCL

CSL

FSL

Predicates C1–C9

Traces + format[1]

Cross-layer graph predicates

S

SCENARIO

Housing subsidy (CSOK)

3 credentials

3 issuers

Cross-credential deps.

Value constraints

G

GOVERNANCE CONSTRAINTS

eIDAS ARF

GDPR 5(1)(c)

W3C VCDM 2.0

Format mandates

Privacy reqs.

Data model conformance

MODEL

DCL

Applicant
Subject

has_children

owns_property

earns

num_children

C4: $\text{area} \geq f(\text{children})$
cross-cred value dep.

property_area

monthly_income

C1: entity alignment
all subjects → same Applicant

C2: trace consistency
each Claim traces to one Prop

trace[1]

trace[1]

trace[1]

CSL

FamilyStatusCred

Civil registry

CS_App₁

num_ch₁

C3 ✓

PropertyCred

Land registry

CS_App₂

prop_a₁

C3 ✓

IncomeCred

Employer

CS_App₃

mon_inc₁

C3 ✓

C5: eIDAS mandate
EU wallets → SD-JWT-VC

format[1]

C6: pred proof req.
GDPR → income needs ZKP

format[1]

C7: VCDM conform.
W3C → AnonCreds fails

format[1]

FSL

SD-JWT-VC

eIDAS ✓

SD-JWT-VC

eIDAS ✓

C9: cross-credential predicate gap
requires cross-cred ZKP; no format supports it

SD-JWT-VC

eIDAS ✓ W3C ✓

pred X

AnonCreds

pred ✓

eIDAS X W3C X

C8: governance conflict
C5∧C6∧C7 jointly unsatisfiable on IncomeCred

eIDAS

GDPR

W3C

RESULT

1

VERIFY

SAT

design is consistent

UNSAT

constraints conflict



2

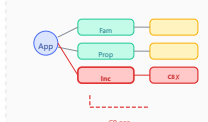
IDENTIFY ERRORS

governance_conflict

(IncomeCred)

cross_cred_pred_gap

(area, children)



3

GENERATE

Relax C6 (GDPR) → 2 valid configurations

Config A



Config B

